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Using Quantum Monte Carlo Simulation to Price Complicated Derivatives in the Big Data Environment

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ABSTRACT

Since the beginning of the 21st century, the scale and complexity of financial derivatives have increased significantly. Meanwhile, the pricing models of derivatives have become increasingly complex, and have involved larger-scale data, posing unprecedented challenges to algorithms and computing speed. Although classical computing can still solve the pricing problem of some complex derivatives by adopting the big data simulation algorithms, its speed has become more and more difficult to meet the needs of real-time computing. In recent years, the emerging quantum computing makes full use of the advantages of quantum superposition to improve the computing speed and accuracy. It has been proven to be an efficient computing mode in big data simulation. This paper takes the pricing of European options and defaultable bonds as examples to explore the application of Quantum Monte Carlo Simulation of complicated derivative pricing in big data. The innovative comprehensive method of pricing derivatives by Quantum Monte Carlo Simulation provides new development for the application of quantum computing in the pricing of complicated financial derivatives (taking defaultable bond as example) with big data simulation. In the case of defaultable bond pricing, the Quantum Amplitude Estimation has been employed to speed up the Monte Carlo Simulation and the price of it has been successfully obtained with at least the same accuracy, which demonstrates that the Quantum Monte Carlo Simulation is a prospective approach to price the complicated derivatives based on the big data since the quantum bits has dramatically increased.

CCS CONCEPTS

• **Computing methodologies** → Modeling and simulation; Simulation types and techniques; Quantum mechanic simulation.

KEYWORDS

Quantum Monte Carlo Simulation, Algorithm, Big Data, Complicated Derivatives

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1 AN INTRODUCTION TO DERIVATIVE PRICING

Financial derivatives are considered to be the most important instruments in the financial industry during the past half century. According to some incomplete statistics, the total global financial derivatives market has approached 1,000 trillion U.S. dollars since 2021. Financial derivatives are probably the best financial instruments available to hedge against certain risks faced by the organizations. They offer to design risk management strategies for the financial firms, and international firms including manufacturing enterprises can also employ them to offer strong risk management tools to secure against any adverse price movements.

A financial derivative is a financial contract signed between a seller and a buyer, and it is valid until its maturity date. The value of the financial derivative is determined by the expiring price or the price process of the underlying asset such as stocks, bonds, currencies, exchange rates, interest rates and price indices. Each financial derivative gives a payoff function so that the buyer can expect to earn from exercising the contract or not. Generally, payoffs value is the function of the price or price process of the underlying asset(s) during the life period of the derivatives. For these derivatives, an important issue is to fairly price them. The famous Black-Scholes-Merton (BSM) model [1][2][3] has successfully priced a large variety of financial derivatives by constructing the risk neutral portfolio and the corresponding partial differential equation that uses a few parameters. Many studies has contributed to improving the Black-Sholes-Merton model to include the more complex payoff structures and more parameters, but it is usually impossible to obtain the closed form solution for many models. Thus, in many cases, the pricing employs Monte Carlo methods that expend vast computational space and time. Therefore, finding more efficient computational algorithm is urgent. In recent years, quantum computation has been widely used in derivative pricing. This article, from an innovative perspective, explores the quantum computing into the process of the derivatives pricing in the context of big data. The computation comparison has demonstrated good

prospect for quantum computing in the pricing of both simple and complicated derivatives on the background of big data.

Prior to introduction of quantum mechanics to computational finance, the BSM model shall be reviewed.

2 BLACK-SCHOLES-MERTON OPTION PRICING FORMULA

There are some assumption for Black-Scholes-Merton pricing formula to hole, and they are as follows:

1. The underlying security prices S follow geometric Brownian motion, that is,$dS = \mu S dt + \sigma S dz$ where the drift rate μ and the volatility rate $\sigma$are constants;

2. The market allows short selling of underlying securities;

3. There isn't any transaction cost or tax, and all underlying securities can be infinitely divided;

4. There isn't any cash income payment for the underlying securities during the life period of the derivatives;

5. There isn't any riskless arbitrage opportunities;

6. Securities trading and the underlying asset price change are both continuous;

7. During the life period of derivative securities, the risk-free interest rate r is constant.

Based on the above assumptions, Black and Scholes obtained the differential equation for the following European call option:

$$\frac{\partial P}{\partial t} + rS\frac{\partial P}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 P}{\partial S^2} = rP \tag{1}$$

Where $r$is the risk-free interest rate, S is the current stock price and P is the option price.

By solving this differential equation with the terminal conditionP_T= $\max(S_T - K, 0)$, Black and Scholes gave the pricing formula for European call options as follows:

$$c = SN(d_1) - Ke^{-r(T-t)}N(d_2) \tag{2}$$

Where $c$is the call option price, and

$$\begin{aligned} d_1 &= \frac{\ln(S/K)+(r+\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \\ d_2 &= \frac{\ln(S/K)+(r-\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t} \end{aligned} \tag{3}$$

(3) is the famous Black-Scholes option pricing formula., and among which, $K$is the strike price; t is the current moment , and T is the expiration time , $\sigma$is the volatility of the stock return, that is, the standard deviation of the rate of return; $N(\cdot)$is the cumulative probability distribution function of the standard normal distribution variable, According to the standard normal distribution function properties, we have $N(-x) = 1 - N(x)$.Therefore for European put options, we can apply a similar derivation to arrive at the theoretical Black-Scholes price as:

$$p = Ke^{-r(T-t)}N(-d_2) - SN(-d_1) \tag{4}$$

3 PRINCIPLES AND ALGORITHM OF CLASSICAL AND QUANTUM MONTE CARLO SIMULATION IN THE SIMPLE OPTION PRICING

In practice, if there are too many parameters to be estimated in the partial differential equation (generally more than 3) or there is no explicit solution of the equation (generally existing in the emerging derivatives structure), the method of Monte Carlo simulation can be used to price derivatives instead of getting the closed form solutions like the BSM formula, but the starting point is that a sampling of the underlying asset expiration date price S_T is calculated Based on the BSM model. To put it simply, according to certain rules, the computer uses a random number generator to generate an infinite number of random data paths (matrix represented as A*B in programming, A path, B time points), and then obtains the expected price by averaging. (Supported by the law of large numbers).

An Introduction to Classical Monte Carlo simulation in Option Pricing

The theory foundation of the Classical Monte Carlo method of option pricing is the risk-neutral pricing: under the risk-neutral measure, the option price at time t can be obtained by discounting the expected value of its payoff at the expiring exercising moment T, i.e., the option price $P_t = E^Q[e^{-r(T-t)}g(S_t, S_{t+1}, \cdots, S_T)]$where$E^Q$ is the mathematical expectation operator under the risk neutral measure, and r is the riskless interest rate, and $g(S_t, S_{t+1}, \cdots, S_T)$ is the expected payoff given the price process of the underlying asset [4]. It is obvious that to price the option is to calculate the expected value, and the Monte Carlo method is employed to estimate the mathematical expectation under the risk neutral measure, and then the option price can be obtained. In general, the Classical Monte Carlo Simulation method for option pricing consists of the following steps (taking European call options as an example):

(l) Simulate the price process (time series) of the underlying security under the risk-neutral measure:

Suppose the properly chosen n time points are$0 = t_0 < t_1 < t_2 < \cdots < t_n = T$, then the time series at these points of the underlying asset price process at the path j is

$$\begin{aligned} &S^j(t_{i+1}) = S^j(t_i)e^{(r-\frac{1}{2}\sigma^2)(t_{i+1}-t_i)+\sigma\sqrt{t_{i+1}-t_i}z_i}, \\ &z_i \sim N(0,1) \end{aligned} \tag{5}$$

(2)At the current time point t, discount the payoff of the option at the maturity moment on this path, with the risk-free interest rate

$$c^j = e^{(-r(T-t))}\max\left\{0, S_T^j - K\right\} \tag{6}$$

(3)Repeat the steps (1) and (2) to obtain a sufficiently large number of sample paths of the discounted valuec^j

(4)Calculate the sample mean to obtain the Classical Monte Carlo simulation value of the option price

$$\begin{aligned} c_t &= \frac{1}{m}\exp(-r(T-t))\sum_{j=1}^{m} c^j \\ &= \frac{e^{(-r(T-t))}\sum_{j=1}^{m}\max\left\{0, S_T^j - K\right\}}{m} \end{aligned} \tag{7}$$

Moreover, it's easy to obtain the probabilistic error boundary between the Classical Monte Carlo Simulation value and the true value. Since$c^j = \mathrm{e}^{(-r(T-t)}\max\{0, S_T^j - K\}$, the mean payoff of the m paths is$\bar{c} = \frac{1}{m}\sum_{i=1}^{m} c^j$, and the variance of m paths is$Var(c) = \frac{1}{m-1}\sum_{i=1}^{m}(c^j - \bar{c})^2$, then the $1-\alpha$ confidence interval is $[\bar{c} - z_{\frac{\alpha}{2}}\sqrt{\frac{\mathrm{var}(c)}{m}}, \bar{c} + z_{\frac{\alpha}{2}}\sqrt{\frac{\mathrm{var}(c)}{m}}]$ where $z_{\alpha/2}$ is the number that

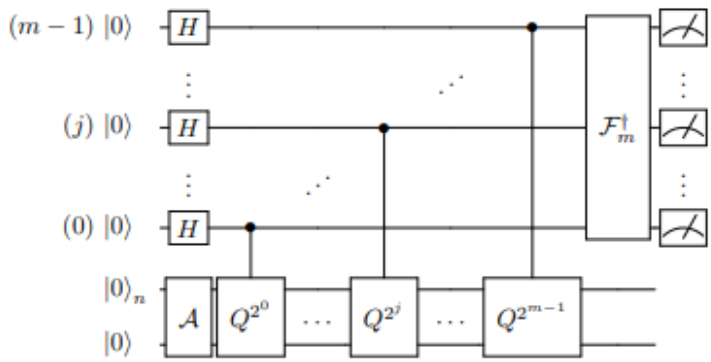

Figure 1: Quantum circuit used to estimate a

the probability that the standard normally distributed variable is greater than it is $_{\alpha/2}$

2. An Introduction to the Quantum Monte Carlo simulation in Option Pricing

Different from the classical Monte Carlo algorithm, the quantum Monte Carlo algorithm uses the mathematical expectation mean and standard deviation of expiry priceS_T to generate the value interval of S_T, that is, the interval of 3 times that the standard deviation below and above the mathematical expectation, then directly sample the $S_T$with identical steps on this interval, the number of samples is$2^n$where n is the number of qubits that store the samples. These samples are lognormally distributed, so the probability of occurrence of each sample can be calculated. At this time, the sample values can be mapped to the 2^nnumbers represented by the n qubits, and all the probabilities can be stored in this group of qubits as the amplitudes of these numbers.

Next, apply the quantum computing to this group of qubits to calculate the payoff value of eachS_T, which is actually a very simple piecewise linear function$Max(S_T - K, 0)$. But the principle of quantum computer is different from that of the traditional computers. Its realization of piecewise linear function is not intuitive. It adds multiple auxiliary qubits to the original qubit group, and then builds a quantum circuit, and the circuit acts on all these qubits, which can be mathematically understood as constructing a specific matrix, multiplied by the matrix composed of the vectors represented by all qubits, the basic procedures will be described later in this article, and the quantum circuit, which implements a piecewise linear function, stores the mathematical expectation of payoff estimated with a certain precision in the amplitude of an auxiliary bit. In fact, at this point the desired option pricing has been obtained, but we can't read it out. If we measure it directly, we need an infinite number of measurements (each measurement is at least one matrix budget) to know what the amplitude represents probability, so the gain outweighs the loss.

Finally, we use the Quantum Amplitude Estimation algorithm to extract the amplitude of this auxiliary qubit. The Quantum Amplitude Estimation algorithm is the last step in almost all quantum algorithm application scenarios, because we need to get the numbers from the quantum world back to the traditional computer world.

Quantum Amplitude Estimation (QAE) algorithm was proposed by Gilles Brassard et al. [5][6],and was advanced in Monte Carlo Simulation by some researchers[[7][8][9]. It is a quantum parameter estimation algorithm, and its main idea is to transform the parameter estimation problem into an operator operation, and map the parameters to the corresponding quantum state amplitudes, and the corresponding parameter estimates are then given by the measurement. According to Nikitas Stamatopoulos et al. [10], QAE has a quadratic level of convergence acceleration compared with Monte Carlo Stochastic Simulation Algorithm operator $\mathcal{A}$, which transforms the quantum state as follow

$$\begin{aligned}\mathcal{A}|\,0>_{n+1} = \sqrt{1-a}\,|\psi_0>_n|0> \\ +\sqrt{a}|\psi_1>_n|1>\end{aligned} \tag{8}$$

Where a is the parameter to be estimated, and the estimate of$a$can be given by the phase estimation algorithm, and its quantum circuit is shown in Figure 1.

Where $\mathcal{F}_m^+$ is the inverse quantum Fourier transform of m bits, and Q is the controlled operation gate, according to Gilles Brassard et al. Q is of the form

$$Q = \mathcal{A}S_0\mathcal{A}^+S_{\psi_0} \tag{9}$$

Where $S_0 = 1 - 2|0><0|$, $S_{\psi_0} = 1 - 2|\psi_0>|0><\psi_0|<0|$

Q is equivalent to a revolving door. Calculated by quantum phase estimation in Figure 1, the method can give an estimate ofθ_a:

$$\widehat{\theta}_a = \frac{y\pi}{M} \tag{10}$$

Where y is the measured value using the inverse quantum Fourier transform and M is the number of quantum samples. Derived by Gilles Brassard[6], the estimate of parameter$a$ is

$$\hat{a} = \sin^2(\hat{\theta}_a) = \sin^2(\frac{y\pi}{M}) \tag{11}$$

It is not difficult to prove that the convergence rate of the parameters estimated by quantum QAE

$O(\frac{1}{M})$, i.e. , $|a - \hat{a}| \leq \frac{\pi}{M} + \frac{\pi^2}{M^2} = O(\frac{1}{M})$ Whereas the convergence rate of the algorithm estimated using Classical Monte Carlo

Table 1: Comparison of the pricing outcomes

Method	**The option price**
BSM Model	4.1905
Classical Monte Carlo (n=1000000)	4.1896
Quantum Monte Carlo (IBM Quantum Experience)	4.1911

Simulation is$O(\frac{1}{\sqrt{M}})$, obviously the QAE algorithm has quadratic level of convergence speedup than MC

(a)Preparation for the quantum state of the underlying asset price (the stock price)

It is now assumed that the maturity price of the underlying asset follows a log-normal random distribution, we need to construct a quantum circuit first, load this random distribution into a quantum state to represent S_T. The distribution will be truncated to the given interval$[S_{T_{\min}}, S_{T_{\max}}]$, and use $2^n$for mesh discretization (n is the numbers of qubits) to represent all of the price intervals$\{0, 1, 2, \cdots 2^{n-1}\}$.

The superposition of quantum states can be obtained by the following transformation:

$$|0>_n \rightarrow |\psi>_n = \sum_{i=0}^{2^n-1} \sqrt{p_i}\,|i>_n \tag{12}$$

Where $\mathrm{p_i}$ represents the probability of the quantum state$i>_n$, and compress i as follows

$$\begin{aligned}&\left\{0, 1, 2, \cdots, 2^{\mathrm{n}-1}\right\} \in i \rightarrow \frac{S_{T_{\max}} - S_{T_{\min}}}{2^n - 1}\\ &\times i + low \in \left[S_{T_{\min}}, S_{T_{\max}}\right]\end{aligned} \tag{13}$$

(b) Preparation for the quantum state of the derivative price (the option price)

As mentioned above, when a European call option expires, if the underlying asset price $S_T$is less

than K, the value of the payoff function is 0, and if $S_T$is greater than K, the value will linearly goes up to infinity. This behavior can be achieved by quantum comparators. Flips the qubit from$|0>$ to$|1>$ when $S_T$is greater than$K$, it can control the linear part of the payoff function, where the price of the European call option can be expressed in quantum state as:

$$\begin{aligned}&|\psi>_n\ |0> \rightarrow |\phi_1\ >\\ &= \sum_{i<K} \sqrt{p_i}\,|i>_n\ |0\ > + \sum_{i\geq K} \sqrt{p_i}\,|i>_n\ |1\ >\end{aligned} \tag{14}$$

Combined with the linear function operator with the QAE algorithm, the amplitude can be estimated

$$\hat{a} \approx \frac{1}{2} - c + \frac{2c}{i_{\max} - K} \sum_{i\geq K} p_i(i-k) \tag{15}$$

Then, the fair expected payoff can be simulated through this quantum states of the option, finally the European call option price can be calculated after with the simulated expected payoff being discounted with the risk free interest rate.

Generally, the Quantum Monte Carlo option price simulation is mainly divided into three steps: firstly prepare the probability distribution of superposed quantum states, and then realize the profit function through the quantum circuits, and finally extract the financial value of the derivative prices through the Quantum Amplitude Estimation and conduct the simulation.

The quantum circuit construction in this article is based on the IBM Quantum Experience [9].

An empirical example is as follows:

Parameter values: $S_t = 6, K = 2, r = 0.1, \sigma = 0.25, T - t = 1$

Then the price calculated using the 3 method is as the following Table 1

It can be seen that the pricing results with Quantum Monte Carlo simulation is closer to the accurate result (with BSM Model) than the Classical Monte Carlo simulation. And most other test also verify the conclusion.

In addition, the European call option price interval obtained by the Quantum Monte Carlo Simulation at the 95% confidence is [4.1886, 4.1922], whereas the 95% confidence level interval by the Classical Monte Carlo simulation is [4.1884, 4.1944].

Examples of plain vanilla option suggests that the pricing results by Quantum Monte Carlo simulation are at least as accurate as that by Classical Monte Carlo Simulation, so how about complex derivatives?

The following section will demonstrate the trial outcome of the defualtable bonds.

4 COMPARISON OF PRICING THE DEFAULTABLE BOND USING THE CLASSICAL AND QUANTUM MONTE CARLO SIMULATION IN THE BIG DATA ENVIRONMENT

As for the defaultable bond, which is representative of complicated derivatives, there are two pricing approaches: structural form and reduced form. Merton proposed the structural form approach and suggested that there must be default if the asset value is less than that of the liability on the expiring date, therefore the default bond is actually the portfolio of long position of the default-free bond and short position of the put option with strike price equal to the total liability value on the expiring date. i.e., the default bond is a type of complicated derivative.

In the next section, the asset value is assumed to follow the Asymmetrically Displaced Double Gamma (AD-DG) Jump-Diffusion Process[11] [12], by which the structural form approach has been extended, the pricing of the defaultable bond will be conducted with the quantum Monte Carlo simulation compared with the classical Monte Carlo simulation on the background of big data..

The problem of bond pricing is different from that of the simple option with fixed strike price for the reason that the asset and the liability of the same corporation are both uncertain and inter

-correlated, thus to build the bond pricing models specially is necessary. As for the AD-DG Jump-Diffusion Process, he pricing model has been modified to conform to the real case in the market by Gen Chen [12] who extends the shape parameters from non-negative integers to non-negative real numbers.

In the extended model [12], the standard Brownian Motions in the asset process$B_A(t)$ and that in the liability $B_D(t)$ are correlated as$E[dB_A(t) \cdot dB_D(t)] = \rho dt$,

The liability value follows the Ito process

$$dD_t/D_t = rdt + \sigma dB_D(t) \tag{16}$$

The asset value follows the extended (AD-DG) Jump-Diffusion Process

$$dA_t/A_t^- = (\mu - \lambda\theta)dt + \sigma_A dB_A(t) + (J-1)dN(t) \tag{17}$$

Denote the logarithm of the jump magnitude of the asset return plus 1 by Z, and assume that Z follows the Asymmetrically Displaced Double Gamma distribution.

$$\begin{aligned} f_Z(x) &= \frac{p\eta_1^{\delta_1}}{\Gamma(\delta_1)}(x-\kappa_1)^{\delta_1-1}e^{-\eta_1(x-\kappa_1)}1_{\{x\geq\kappa_1\}} \\ &+\frac{(1-p)\eta_2^{\delta_2}}{\Gamma(\delta_2)}(\kappa_2-x)^{\delta_2-1}e^{\eta_2(x-\kappa_2)}1_{\{x\leq\kappa_2\}} \end{aligned} \tag{18}$$

Through the Esscher transform, the risk-neutral AD-DG Jump-Diffusion Process of the asset-liability ratio $X_t := A_t/D_t$with the new parameters can be obtained for the simulation with quantum Monte Carlo Simulation.

Since there are 18 parameters to be input, the Quantum Monte Carlo Simulation can be conducted with the speed no lower than the Classical Monte Carlo Simulation proposed in [12] in the big data environment.

Step 1: Input 18 parameters

Step 2: Calculate the transitional variables

Step3: Build the quantum bits as **the part 2 of section III in this article and simulate the sum of the jump magnitude as follows**

$$\begin{aligned} &\sum_{m=1}^{N(t)} Z_m \sim [\Gamma(n(t)\delta_1,\eta_1) + n(t)\kappa_1] \\ &-[\Gamma((N(t)-n(t))\delta_2,\eta_2) - (N(t)-n(t)\kappa_2] \\ &= \Gamma(n(t)\delta_1,\eta_1) - \Gamma((N(t)-n(t))\delta_2,\eta_2) \\ &+n(t)(\kappa_1-\kappa_2) + N(t)\kappa_2 \end{aligned} \tag{19}$$

Using the same parameters by Gen Chen [12] and 127qubits, the Quantum Monte Carlo Simulation of the defaultable bond price P_Q=.0.6l831with, which is quite close to the Classical Monte Carlo Price (P_C=0.61821). Moreover, the 95% credit interval for the Quantum Monte Carlo Simulation of the defaultable bond price is [0.59987, 0.61972] compared with the 95% credit interval [0.58897, 0.62012] for the classical simulation of the price. As for the Quantum Monte Carlo Simulation, there are 3 steps, and the circuit is similar to that of vanilla option pricing as explored in part III of this article. But the speed-up is obvious compared with the classical Monte Carlo Simulation, especially for the AD-DG model applied in the defualtable bond pricing, which involves many parameters and it can be only dealt with in the environment of big data. The classical Monte Carlo simulation can also price the complicated derivatives, however, the quantum Monte Carlo simulation is more hopeful to deal with the big data to price them in a relatively short period of time with at least with the same precision [12].

5 CONCLUSION

Based on the comparison above, it can be seen that the Quantum Monte Carlo Simulation has obtained the price of the defaultable bond (belonging to complicated derivatives) which is no less accurate than that of the Classical Monte Carlo Simulation even if the former has used only 127 qubits. In the big data environment, many parameters, vast amount of data, high speed and real time computing is in increased need by the more and more complicated derivative pricing problems, thus the fast growth of the qubits has become the strong support of the Quantum Monte Carlo Simulation in pricing the complicated derivatives. On Nov. 9th 2022, IBM declared the Osprey chip with 433 qubits, which will greatly boost the computing efficiency, and it has planned to launch the Quantum System 2 with 1000qubits late in 2023, thus the current Quantum Monte Carlo Simulation approach for derivative pricing will obtain revolutionary improvement [13] [14] [15].

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